Primary Cloud

PUT (presigned)

End User

login

Cognito
User Pool + Hosted UI
ID token issurance

API call + ID token

API Gateway
REST + Cognito Authorizer
CORS enabled

invoke

Upload URL Lambda
POST /upload handler
S3 presigned PUT URL generator
Content-type allowlist (img/video)

presigned URL

write

Upload Pipeline
Lambda Ingest + Dedup (SHA256)
Lambda Thumbnail (OpenCV)
Lambda Frame Extract (ffmpeg, 1 fps)

ObjectCreated trigger

lambda.invoke (async)

Tagging (Container)
ECR Image · MegaDetector + SpeciesNet (46 species)
model_uri env var → S3

PutItem (file_id, tags, URLs)

invoke

Query & Mgmt
Lambda Query × 4 (AND logic)
Lambda Tag Add/Remove (bulk)
Lambda Delete (storage + DB)
endpoints: /query/{tags|species|thumbnail|file}, /tags/edit, /delete

invoke

Notifications Subscribe
POST /notifications/subscribe
Creates SNS HTTPS sub w/?email=&filter policy

sns.subscribe()

S3 Buckets
raw-uploads
thumbnails
video-frames
models (.pt)
query-tmp (TTL)
web-static (UI)

read / delete

DynamoDB
a2_group92_files
PK: file_id
GSI: checksum_sha256-index
a2_group92_tag_index
PK: tag, SK: file_id

Query GSI / GetItem / UpdateItem

publish (String.Array tags)

SNS Topic
tag filter policy
(per-user subs)

S3 copy (cross-cloud)

HTTPS push (cross-cloud)

Secondary Cloud

Google Cloud Platform

Secondary Storage

Cloud Run
S3 → GCS copy bridge
X-Copy-Secret auth

GCS Bucket
Mirror of S3 media
originals + thumbs + frames

Notifications

Cloud Function
SNS HTTPS push receiver
?email=<user> in URL

Resend
HTML email API
3000/mo free, no domain auth

deliver email

Email Inbox

IAM Roles (Academy Constraint)

All Lambdas use LabRole (preset). Fine-grained access via S3 bucket policies +DynamoDB resource policies. Cognito ID token verified at API Gateway authorizer.All endpoints require valid JWT except /signup
API Gateway uses Cognito Authorizer; Lambdas read claims from event.requestContext.authorizer.claims.